Architecture-dependent effects of imbalance correction in melanoma classification: a controlled audit with fairness, calibration, and explanation analysis

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Abstract

Melanoma classifiers are trained on imbalanced, predominantly light-skinned data. We ask whether imbalance correction changes CNN and transformer performance, whether calibration explains it, and how the models behave across skin tone and lesion contrast. We benchmark ResNet-50, EfficientNet-B0, and ViT-S/16 on HAM10000 under three pre-declared conditions: standard cross-entropy, focal loss with class-balanced sampling, and cross-entropy with strong augmentation. Each condition uses three seeds, a lesion-aware split, and a validation-selected threshold at 95% specificity. Focal loss raises ViT-S/16 sensitivity by 8.2 points and lowers ResNet-50 by 11.7 and EfficientNet-B0 by 7.9. The direction is consistent, but every within-architecture interval includes zero. The EfficientNet-B0 versus ViT interaction is significant; the ResNet-50 versus ViT interaction is not. The AUC ordering is preserved, but the ViT closes the gap under focal loss, so the reversal is not purely a threshold effect. Calibration error rises 3.0x to 4.4x for the CNNs and 1.36x for the ViT, so calibration tracks the CNN loss, not the ViT gain. The dark stratum has a higher false positive rate at L^* cutoffs of 55, 58, and 60, but the dark AUC differences are not significant at any cutoff. The false-negative L^* association is significant for all three architectures; the false-positive association is significant only for ResNet-50. Grad-CAM mass is lower on darker images, but its error relationship differs by architecture. HIBA validation shows consistent CNN losses under focal loss, a seed-dependent ViT gain, and the best external performance under strong augmentation. This is a controlled audit.

Keywords: melanoma classification, class imbalance, fairness, calibration, explainable artificial intelligence, skin tone

1 Introduction

Melanoma is the most lethal form of skin cancer, and survival depends on early detection. Deep learning models have reached dermatologist-level accuracy on public dermoscopic benchmarks such as HAM10000 and the ISIC challenges [1, 2]. Three problems remain.

Public datasets are imbalanced. Benign lesions outnumber melanomas by roughly ten to one, and a classifier that predicts the majority class can look accurate while missing cancers. Focal loss and class-balanced sampling are common remedies [3], but their effect is often reported for one architecture and one dataset.

The datasets are also skewed toward lighter skin. Fitzpatrick skin types V and VI are rare in public dermoscopic data, and models trained on lighter skin can lose sensitivity on darker skin [4, 5, 6]. The data needed to measure the gap are themselves scarce.

Deep models are opaque as well. Gradient-based attribution methods such as Grad-CAM and Integrated Gradients can show which pixels drive a prediction [7, 8], but most dermatology papers use them as illustrations. They are rarely turned into quantitative measurements that can be tested across skin tone groups.

Most studies examine one of these problems at a time. Architecture comparisons often apply a single imbalance correction without checking whether it is neutral across architectures, and fairness studies report group-level performance without connecting it to attribution behaviour. This study connects the three in one benchmark.

We study HAM10000 [1] with a binary melanoma endpoint and three architectures: ResNet-50, EfficientNet-B0, and ViT-S/16. We use a lesion-aware split, three seeds per condition, and a validation-selected threshold at 95% specificity. The test set is used for evaluation only and is not used for model selection.

Our contributions are:

1. A controlled benchmark of three architectures under three pre-declared conditions: CE baseline, focal loss with class-balanced sampling, and CE with strong augmentation.

2. Evidence that focal loss with class-balanced sampling changes the sensitivity of all three architectures in a consistent direction across seeds, while every within-architecture 95% t interval includes zero. The EfficientNet-B0 versus ViT interaction is significant at the seed level; the ResNet-50 versus ViT interaction is not. The AUC ordering is preserved, but the ViT closes the gap under focal loss.

3. A calibration analysis showing that focal loss raises expected calibration error by 3.0x to 4.4x for the CNNs and 1.36x for the ViT. Calibration tracks the CNN loss, not the ViT gain.

4. A lightness-stratified fairness analysis at three cutoffs. The dark stratum has a higher false positive rate at every cutoff, but the dark AUC differences are not significant at any cutoff. The false-negative L^* association is significant for all three architectures; the false-positive association is significant only for ResNet-50.

5. A quantitative XAI analysis with Grad-CAM targeting the true class. All three architectures place less attribution inside the lesion on darker images, but the mass-error relationship differs by architecture and the false-negative versus false-positive mass values do not separate consistently. The results are reported in Section 4.6.

6. An external validation on HIBA showing a consistent CNN sensitivity loss under focal loss and a seed-dependent ViT gain, with no Fitzpatrick IV melanomas and no Fitzpatrick V-VI images.

The training and fairness analysis use one public dataset, HAM10000. The lightness strata are image-based measurements, not clinical Fitzpatrick labels, and the results do not establish clinical utility. External validation on HIBA is reported in Section 4.7.

2 Related work

2.1 Melanoma classification and public datasets

Deep learning for skin lesion classification moved from dermatologist-level comparisons on single datasets [9] to multi-source benchmarks. HAM10000 contributed 10,015 dermoscopic images across seven diagnostic classes [1], and the ISIC challenges expanded the scale and the number of sources [2]. BCN20000 added 18,946 dermoscopic images with eight diagnostic classes [10], and the ISIC 2020 collection added patient-level context and a clinical endpoint [11]. Cassidy et al. analysed the ISIC datasets and showed that duplicate lesions, class imbalance, and inconsistent metadata affect benchmark comparisons [12]. Human-reader studies and reporting guidelines now set a higher bar for evaluation [13, 14]. Clinical translation also requires prospective validation and reporting standards that go beyond a single benchmark [15].

Architecture comparisons usually cover a ResNet-style CNN [16], an efficient CNN [17], and a vision transformer [18]. The comparisons differ in preprocessing, split protocol, and threshold choice, which makes cross-paper rankings difficult to interpret. This paper fixes the split, the seeds, and the operating point so that the architecture and the training condition are the only variables.

2.2 Class imbalance

Focal loss down-weights easy examples and is the most common loss-level remedy for class imbalance [3]. Class-balanced loss reweights classes by their effective number of samples [19], and class-balanced sampling changes the effective class prior directly. Johnson and Khoshgoftaar survey the broader set of resampling, reweighting, and augmentation methods [20]. Controlled studies in convolutional networks show that the best correction depends on the architecture and the dataset size [21]. Strong photometric and geometric augmentation is often added at the same time.

The three interventions are usually evaluated together, so their separate effects and their interactions with architecture are rarely reported. This matters because a correction that is neutral on average may help one architecture and hurt another. The present benchmark treats the correction as a pre-declared condition and tests the interaction directly.

2.3 Fairness in dermatology

Fitzpatrick17k provided the first large clinical image set with Fitzpatrick labels and showed performance gaps across skin types [4]. The Diverse Dermatology Images dataset added biopsy-proven clinical images with balanced skin tone representation [5]. SCIN added crowdsourced clinical images with self-reported and estimated Fitzpatrick labels [22]. Pipsqueak collected dermoscopic melanocytic lesions in darker skin and reported a large sensitivity drop on Fitzpatrick V-VI [6]. A recent review identifies fairness, generalizability, and replicability as the main unresolved problems in melanoma AI [23].

The dermoscopic datasets with Fitzpatrick labels remain small. HIBA contains 1,616 images from Argentina with Fitzpatrick labels [24], and PROVe-AI contains 603 biopsy-confirmed lesions [25]. Neither includes Fitzpatrick V-VI. PASSION provides clinical images from Sub-Saharan Africa with Fitzpatrick III-VI [26], but the imaging modality differs from dermoscopy. The available data therefore support an image-based lightness analysis on dermoscopy and a separate cross-modality analysis on clinical images, not a single clean Fitzpatrick comparison.

Group fairness also depends on which error is counted. Equal opportunity compares true positive rates across groups [27], while a false positive rate comparison captures the harm of unnecessary

biopsies. A model can improve one and worsen the other. This paper reports both and models them separately.

Shortcut learning complicates the interpretation of group gaps. Surgical skin markings can change melanoma predictions [28], and Bissoto et al. showed that several visual artifacts are exploitable and that current debiasing methods do not remove them [29]. Lesion-skin contrast is one such visual factor: it correlates with skin tone but also with lesion pigmentation. A fairness analysis that does not adjust for it can attribute a contrast effect to skin tone. This paper measures contrast from the lesion masks and includes it in the adjusted models.

2.4 Calibration

Thresholded metrics such as sensitivity at a fixed specificity depend on the score distribution, not only on the ranking quality. Modern neural networks are often miscalibrated [30], and calibration is central to clinical prediction models [31]. A correction that changes calibration can therefore change the sensitivity ranking while leaving the AUC ranking stable. This is the mechanism we test, rather than assume.

2.5 Explainable AI in dermatology

Class activation maps and Grad-CAM are the most used attribution methods in dermatology [32, 7]. Integrated Gradients and other axiomatic methods provide an alternative [8], and model-agnostic methods such as LIME and SHAP are used when gradients are unavailable [33, 34]. Clinician-facing studies show that explanations can improve trust and diagnostic accuracy [35], but the explanations are usually evaluated qualitatively.

Qualitative evaluation is not enough for a fairness claim. A heatmap that looks plausible can be insensitive to the model parameters [36], and explainable AI in health care has been criticized for promising more than it delivers [37]. This paper treats attribution as a measurement with a stated error type, not as evidence of clinical reasoning.

2.6 Quantitative XAI metrics

Attribution localization can be measured against lesion segmentation masks: the fraction of attribution mass inside the lesion, the pointing game, and the attribution IoU. Faithfulness can be measured with deletion and insertion curves [38], sensitivity-n [39], and the ROAD benchmark [40]. These metrics make it possible to test whether an explanation is clinically meaningful rather than visually plausible.

Dermatology studies rarely combine these metrics with skin tone stratification. The XAI analysis in this paper uses localization metrics across tone groups and error types, and it reports why faithfulness curves were not informative for these highly confident models.

2.7 Gap

Three questions remain open: whether imbalance correction changes the ranking of architecture families, whether calibration explains the change, and whether attribution metrics can detect a group-level problem before it appears in aggregate sensitivity. This paper addresses the first two with paired tests and calibration, and it addresses the third with tone-stratified localization metrics and a false-positive/false-negative split.

3 Methods

3.1 Data and splits

We use HAM10000, which contains 10,015 dermoscopic images from 7,470 lesions across seven diagnostic classes [1]. The task is binary: melanoma versus all other diagnoses. The dataset contains 1,113 melanoma images, 11.1% of the total.

We use a lesion-aware split so that images of the same lesion never appear in more than one split. The split is 70% training, 15% validation, and 15% test. The test set contains 1,527 images and 186 melanomas. The split is generated with seed 42, released as `results/split_ham10000_lesion_seed42.csv`, and reused by every run.

3.2 Preprocessing and augmentation

Images are resized to 160 pixels for the Stage 1 screen and 224 pixels for the Stage 2 comparison, and normalized with ImageNet statistics. The standard augmentation pipeline applies horizontal and vertical flips, random 90-degree rotations, and mild color jitter. The strong pipeline increases the color jitter and adds affine transformations with scale, translation, and rotation. The strong regime is designed to simulate domain shift.

3.3 Architectures

We compare ResNet-50, EfficientNet-B0, and ViT-S/16. All models use ImageNet pretrained weights and a two-class output. The ViT uses dynamic image sizing so that it accepts the 224-pixel input. The three models cover a standard CNN, an efficient CNN, and a transformer.

3.4 Training

All models use AdamW with a learning rate of 3×10^{-4}, weight decay 1×10^{-4}, and a cosine schedule. The batch size is 64, the maximum number of epochs is 15, and early stopping uses a patience of 5 on validation balanced accuracy. Gradients are clipped to a norm of 1.0. Training uses mixed precision when a GPU is available.

3.5 Conditions

Stage 1 is a screen on EfficientNet-B0 at 160 pixels. It contains eight configurations: CE with a random sampler, CE with class-balanced sampling, weighted CE with a random sampler, focal loss with $\gamma = 2$ and a random sampler, focal loss with $\gamma = 2$ and class-balanced sampling, focal loss with $\gamma = 3$ and class-balanced sampling, and two strong-augmentation variants. Each configuration uses three seeds, for 24 runs.

Stage 2 compares three pre-declared conditions across the three architectures at 224 pixels:

1. CE baseline: cross-entropy loss, random sampler, standard augmentation.
2. Planned imbalance correction: focal loss with $\gamma = 2$, class-balanced sampling, standard augmentation.
3. Stage 1 winner: cross-entropy loss, random sampler, strong augmentation.

Each condition uses three seeds, for 27 runs. The Stage 2 conditions are fixed before the test results are seen.

3.6 Metrics and threshold

We report sensitivity, specificity, balanced accuracy, AUC, and sensitivity at 95% specificity. The operating threshold is selected on the validation split at 95% specificity, frozen, and applied once to the test split. Each run writes per-sample test predictions so that the same scores can be reused for fairness and XAI analysis.

3.7 Statistical analysis

We report the mean and standard deviation across three seeds. The seed is the experimental unit for a training intervention. We report 95% t intervals across seeds for the mean and release per-seed values in the result files. The intervals are wide with three seeds, and we treat them as descriptive. Image-level bootstrap intervals are not used as confidence intervals for the seed-level mean. The test set is used for evaluation only.

We test AUC differences between conditions and architectures with paired seed-level t intervals. We also compute sensitivity at a common 95% test specificity for every configuration. These analyses separate test-set uncertainty from run-to-run variation.

3.8 Fairness analysis

HAM10000 has no Fitzpatrick labels. We therefore compute an image-based lightness proxy from non-lesional skin. The image is converted to CIELAB, the lesion mask is dilated by 11 pixels and excluded, and very dark pixels ($L^* < 10$) and very bright pixels ($L^* > 95$) are removed. The median L^* of the remaining skin pixels is the proxy.

We group images into light ($L^* > 70$), medium ($55 < L^* \leq 70$), and dark ($L^* \leq 55$). These thresholds are specific to the HAM10000 lightness distribution. They are image-based strata, not clinical Fitzpatrick labels.

We also computed the Individual Typology Angle (ITA), $\text{ITA} = \arctan((L^* - 50)/b^*) \times 180/\pi$. The ITA was unstable on 37% of images because the b^* channel was near zero or negative after dermatoscope color casts. The L^* proxy is therefore the primary analysis, and the ITA is not used for group comparisons.

For each Stage 2 run, we compute per-group sensitivity, specificity, false positive rate, false negative rate, and AUC. Group estimates are reported as mean and standard deviation across seeds.

Lesion contrast is computed from the same masks. The lesion color is the median CIELAB value inside the mask, and the surrounding skin is the median inside an 11-pixel ring after excluding $L^* < 10$ and $L^* > 95$. We define ΔL^* as skin minus lesion, so a positive value means the lesion is darker than the surrounding skin, and ΔE as the CIELAB 76 difference between the two median colors.

The main analysis uses the pre-declared cutoff $L^* \leq 55$. We repeat the group analysis at $L^* \leq 58$ and $L^* \leq 60$ to test whether the results depend on the cutoff. We also fit adjusted logistic models of false positive and false negative on perilesional L^*, ΔL^*, log lesion area, age, sex, and anatomical site. Continuous predictors are standardized; one standard deviation is 7.2 L^* units and 10.3 contrast units in the test set. Standard errors come from a cluster bootstrap over lesions with 300 replicates, so repeated images and seeds of the same lesion are not treated as independent. The p-values are descriptive and are not corrected for multiplicity across cutoffs and architectures.

We compute expected calibration error with 10 equal-width bins and the Brier score on the test split. Paired sensitivity and AUC comparisons are computed per seed and summarized with 95% t intervals across three seeds. Difference-in-differences compares the CE-to-focal change of

one architecture with that of another. The direction count reports how many seeds agree with the mean sign.

External validation uses HIBA [24], a dermoscopic dataset from Argentina with Fitzpatrick labels. We apply the frozen HAM10000 validation threshold and do not refit on HIBA. As a secondary comparison, we also report sensitivity at 95% specificity with a threshold fitted on HIBA. This oracle threshold is an upper bound, not a deployable result. PROVe-AI [25] is a possible extension but was not evaluated here. A clinical dark-skin dataset, such as DDI [5] or PASSION [26], is left for future work, with the imaging modality difference stated as a limitation.

3.9 XAI analysis

For the CE-baseline run of each architecture, we compute Grad-CAM heatmaps on a tone-stratified test sample: all 43 dark images plus 150 randomly drawn light and 150 medium images per seed. Grad-CAM targets the true class for every image, including false negatives, so the heatmap describes evidence for melanoma rather than the predicted class. We measure:

- Attribution mass inside the lesion: $M = \sum_{i \in \text{lesion}} h_i / \sum_i h_i$, where h_i is the normalized heatmap value at pixel i.
- Pointing game accuracy: the fraction of images where the maximum heatmap value falls inside the lesion mask.
- Attribution IoU: the intersection over union between the thresholded heatmap and the lesion mask.

We then compute the Spearman correlation between attribution mass and error status, and between attribution mass and lightness group. The correlations pool the three seeds, so the p-values treat repeated images as independent and are descriptive rather than confirmatory. The XAI analysis uses the same test split and the same per-sample predictions as the classification and fairness analyses.

We also implemented deletion and insertion curves, but the models are highly confident: the target-class probability stays near 1.0 for both the original and the perturbed images, so the curves do not separate heatmaps on this benchmark. We therefore report localization metrics and treat faithfulness curves as future work.

3.10 Reproducibility

Every run writes its configuration, git commit, seed, device, package versions, metrics, and per-sample predictions. The split is cached and reused. Stage 2 training used Kaggle GPUs. The calibration, XAI, and external evaluations used a local Apple Silicon GPU. The code, configs, and table generation scripts are available in the project repository.

4 Results

4.1 Stage 1: loss, sampling, and augmentation screen

Table 1 reports the Stage 1 screen on EfficientNet-B0 at 160 pixels. The validation winner is CE with strong augmentation, with a validation sensitivity of 0.556 ± 0.027 at 95% specificity. The CE baseline reaches 0.536 ± 0.016. The planned focal loss with class-balanced sampling reaches 0.502 ± 0.039, and focal loss with $\gamma = 3$ and class-balanced sampling reaches 0.449 ± 0.051.

Table 1: Stage 1 screen on EfficientNet-B0 at 160 pixels. Validation sensitivity is used for model selection. Validation values are mean and standard deviation across three seeds. Test sensitivity is the mean with a 95% t interval across three seeds at the validation-selected 95% specificity threshold.

Configuration	Val sens	Test sens (95% t CI)	Specificity	Balanced acc	AUC
stage1_ce_random_strong	0.556 ± 0.027	0.513 (0.492 to 0.533)	0.943	0.728	0.901
stage1_ce_random	0.536 ± 0.016	0.530 (0.470 to 0.591)	0.948	0.739	0.897
stage1_ce_cb	0.521 ± 0.086	0.484 (0.403 to 0.565)	0.938	0.711	0.890
stage1_focal2_random	0.509 ± 0.041	0.487 (0.433 to 0.541)	0.946	0.717	0.894
stage1_focal2_cb	0.502 ± 0.039	0.480 (0.361 to 0.600)	0.943	0.712	0.897
stage1_wce_random	0.502 ± 0.043	0.493 (0.430 to 0.556)	0.937	0.715	0.896
stage1_focal2_cb_strong	0.483 ± 0.013	0.491 (0.274 to 0.708)	0.943	0.717	0.896
stage1_focal3_cb	0.449 ± 0.051	0.478 (0.430 to 0.527)	0.946	0.712	0.899

The planned imbalance correction did not beat the CE baseline on validation. The difference between the CE baseline and the strong-augmentation variant is small relative to the seed standard deviation. The test columns in Table 1 are descriptive; the model selection used validation only.

4.2 Stage 2: architecture comparison

Table 2 reports the Stage 2 comparison at 224 pixels. The three conditions are the CE baseline, focal loss with class-balanced sampling, and CE with strong augmentation.

At the CE baseline, ResNet-50 reaches a test sensitivity of 0.570 (95% t interval 0.469 to 0.671), EfficientNet-B0 reaches 0.556 (0.465 to 0.646), and ViT-S/16 reaches 0.400 (0.291 to 0.508). The two CNNs lead the transformer in the point estimates, but the intervals overlap.

Under focal loss with class-balanced sampling, ResNet-50 falls to 0.453 (0.289 to 0.618), EfficientNet-B0 falls to 0.477 (0.295 to 0.658), and ViT-S/16 rises to 0.482 (0.428 to 0.536). The correction changes the point estimates by +8.2 points for the transformer and by −11.7 and −7.9 points for the two CNNs.

Strong augmentation is close to neutral. ResNet-50 reaches 0.568 (0.495 to 0.642), EfficientNet-B0 reaches 0.563 (0.402 to 0.724), and ViT-S/16 reaches 0.435 (0.228 to 0.643).

Table 3 reports the seed-level paired comparisons. Every ResNet-50 and EfficientNet-B0 seed falls under focal loss, and every ViT-S/16 seed rises, but all three within-architecture 95% t intervals include zero. The difference-in-differences is −0.161 (−0.209 to −0.113) for EfficientNet-B0 minus ViT-S/16 and −0.199 (−0.484 to 0.087) for ResNet-50 minus ViT-S/16. The EfficientNet-B0 versus ViT interaction is therefore significant at the seed level; the ResNet-50 versus ViT interaction is directionally consistent but not significant.

The point-estimate ranking reverses, but the evidence supports an architecture-dependent interaction with limited precision, not a definitive ordering. The same caveat applies to the strong-augmentation changes, whose intervals all include zero.

At a common 95% test specificity, the CE CNNs reach 0.581 and 0.575 against 0.348 for the ViT. Under focal loss the values are 0.450, 0.466, and 0.437. The ViT gain and the CNN loss therefore persist at a matched operating point, not only at the validation-selected threshold.

4.3 AUC ordering is preserved while the gaps close

The AUC ordering does not follow the sensitivity ranking. At the CE baseline, the AUC is 0.914 for EfficientNet-B0, 0.897 for ResNet-50, and 0.854 for ViT-S/16. Under focal loss with class-balanced

Table 2: Stage 2 comparison at 224 pixels. Validation values are mean and standard deviation across three seeds. Test sensitivity is the mean with a 95% t interval across three seeds at the validation-selected 95% specificity threshold.

Configuration	Val sens	Test sens (95% t CI)	Specificity	Balanced acc	AUC
stage2_efficientnet_b0_ce_random_strong	0.521 ± 0.021	0.563 (0.402 to 0.724)	0.952	0.757	0.912
stage2_efficientnet_b0_ce_random	0.553 ± 0.035	0.556 (0.465 to 0.646)	0.953	0.754	0.914
stage2_efficientnet_b0_focal2_cb	0.481 ± 0.042	0.477 (0.295 to 0.658)	0.950	0.713	0.891
stage2_resnet50_ce_random	0.526 ± 0.044	0.570 (0.469 to 0.671)	0.951	0.761	0.897
stage2_resnet50_ce_random_strong	0.549 ± 0.015	0.568 (0.495 to 0.642)	0.942	0.755	0.899
stage2_resnet50_focal2_cb	0.489 ± 0.027	0.453 (0.289 to 0.618)	0.947	0.700	0.884
stage2_vit_s_focal2_cb	0.474 ± 0.034	0.482 (0.428 to 0.536)	0.940	0.711	0.872
stage2_vit_s_ce_random_strong	0.466 ± 0.030	0.435 (0.228 to 0.643)	0.941	0.688	0.866
stage2_vit_s_ce_random	0.423 ± 0.042	0.400 (0.291 to 0.508)	0.941	0.670	0.854

Table 3: Seed-level paired sensitivity comparisons. Intervals are 95% t intervals across three seeds. The direction column counts the seeds with the stated sign. The EfficientNet-B0 versus ViT interaction is the only mean difference whose interval excludes zero; the other intervals include zero, so the study is underpowered for a precise ranking.

Test	Architecture	Δ sensitivity	95% t CI	Direction
Focal+CB vs CE	EfficientNet-B0	-0.079	-0.219 to +0.061	3/3 negative
Focal+CB vs CE	ResNet-50	-0.116	-0.272 to +0.039	3/3 negative
Focal+CB vs CE	ViT-S/16	+0.082	-0.079 to +0.244	3/3 positive
CE+strong vs CE	EfficientNet-B0	+0.007	-0.244 to +0.258	2/3 positive
CE+strong vs CE	ResNet-50	-0.002	-0.069 to +0.065	2/3 negative
CE+strong vs CE	ViT-S/16	+0.036	-0.075 to +0.147	2/3 positive
Difference in differences	EfficientNet-B0 minus ViT-S/16	-0.161	-0.209 to -0.113	3/3 negative
Difference in differences	ResNet-50 minus EfficientNet-B0	-0.038	-0.283 to +0.208	1/3 negative
Difference in differences	ResNet-50 minus ViT-S/16	-0.199	-0.484 to +0.087	3/3 negative

sampling, the AUC is 0.891 for EfficientNet-B0, 0.884 for ResNet-50, and 0.872 for ViT-S/16. The point-estimate order is unchanged: EfficientNet-B0 first, ResNet-50 second, ViT-S/16 last.

The gaps are not stable. In-domain AUC differences are reported in Table 4. At the CE baseline, EfficientNet-B0 exceeds ViT-S/16 by 0.060 (0.040 to 0.081) and ResNet-50 exceeds ViT-S/16 by 0.043 (0.002 to 0.084). Under focal loss, the corresponding differences are 0.018 (−0.008 to 0.045) and 0.012 (−0.017 to 0.040), so no architecture pair is significantly separated. Focal loss lowers ResNet-50 AUC by 0.013 (−0.023 to −0.003) and EfficientNet-B0 by 0.023 (−0.047 to 0.000), while the ViT changes by +0.018 (−0.003 to 0.040). The ResNet-50 versus ViT AUC interaction is −0.031 (−0.061 to −0.001).

The sensitivity reversal is therefore not purely a threshold effect. The ViT improves both sensitivity and AUC under focal loss, and the CNN AUC also falls. Calibration and threshold placement remain part of the explanation, but they do not account for the whole pattern.

4.4 Calibration tracks the CNN loss

Table 5 reports test-set calibration. Focal loss with class-balanced sampling raises expected calibration error from 0.046 to 0.203 for ResNet-50 (4.4x), from 0.060 to 0.178 for EfficientNet-B0 (3.0x), and from 0.146 to 0.199 for ViT-S/16 (1.36x). Strong augmentation keeps ECE between 0.046 and 0.058. The Brier score follows the same pattern.

Table 4: AUC differences at the CE baseline and under focal loss. Intervals are 95% t intervals across three seeds. Positive values mean the first architecture has the higher AUC. The AUC ordering is preserved at the point-estimate level, but the gaps are not significant under focal loss.

Condition	Comparison	Δ AUC	95% t CI
CE	EfficientNet-B0 minus ViT-S/16	+0.060	+0.040 to +0.081
CE	ResNet-50 minus EfficientNet-B0	-0.017	-0.044 to +0.010
CE	ResNet-50 minus ViT-S/16	+0.043	+0.002 to +0.084
Focal+CB	EfficientNet-B0 minus ViT-S/16	+0.018	-0.008 to +0.045
Focal+CB	ResNet-50 minus EfficientNet-B0	-0.007	-0.036 to +0.022
Focal+CB	ResNet-50 minus ViT-S/16	+0.012	-0.017 to +0.040

Table 5: Test-set calibration. ECE uses 10 equal-width bins and is independent of the operating threshold. Values are mean and standard deviation across three seeds.

Configuration	ECE	Brier
ResNet-50 / CE	0.046 ± 0.013	0.074 ± 0.004
ResNet-50 / CE+strong	0.046 ± 0.004	0.076 ± 0.004
ResNet-50 / Focal+CB	0.203 ± 0.041	0.133 ± 0.016
EfficientNet-B0 / CE	0.060 ± 0.009	0.076 ± 0.001
EfficientNet-B0 / CE+strong	0.052 ± 0.015	0.076 ± 0.003
EfficientNet-B0 / Focal+CB	0.178 ± 0.049	0.134 ± 0.022
ViT-S/16 / CE	0.146 ± 0.109	0.126 ± 0.041
ViT-S/16 / CE+strong	0.058 ± 0.037	0.089 ± 0.012
ViT-S/16 / Focal+CB	0.199 ± 0.039	0.141 ± 0.011

The calibration change tracks the CNN sensitivity loss, but it does not explain the ViT gain. The ViT gains sensitivity and AUC while its ECE rises only modestly. The CNNs lose both sensitivity and calibration. Calibration is therefore part of the mechanism for the CNN degradation, not a single explanation for the ranking reversal.

Figure 1 shows reliability curves and ECE by lightness group. The dark stratum has higher ECE than the light and medium groups for all three CE baselines: 0.126 against 0.072 and 0.052 for EfficientNet-B0, 0.132 against 0.063 and 0.035 for ResNet-50, and 0.235 against 0.144 and 0.149 for ViT-S/16.

4.5 Fairness across lightness strata

The test set contains 644 light images with 81 melanomas, 840 medium images with 99 melanomas, and 43 dark images with 6 melanomas at the pre-declared cutoff $L^* \leq 55$. The dark stratum is 2.8% of the test set and 3.2% of the melanoma cases.

Table 6 reports the per-group sensitivity at 95% specificity for the nine Stage 2 configurations at the pre-declared cutoff. At this cutoff the dark stratum has the lowest AUC in all nine configurations, 0.71 to 0.87, against 0.83 to 0.90 for light and 0.88 to 0.93 for medium. It also has the highest false positive rate, 0.08 to 0.19, against 0.04 to 0.07 for the other two groups.

The dark AUC differences are not statistically supported. Across the three cutoffs, none of the 54 dark-versus-light and dark-versus-medium AUC comparisons has a bootstrap interval that excludes zero. The dark AUC deficit is therefore a descriptive feature of the strict cutoff, not a confirmed group difference. The false positive rate is more consistent: it is 0.126 at $L^* \leq 55$, 0.086

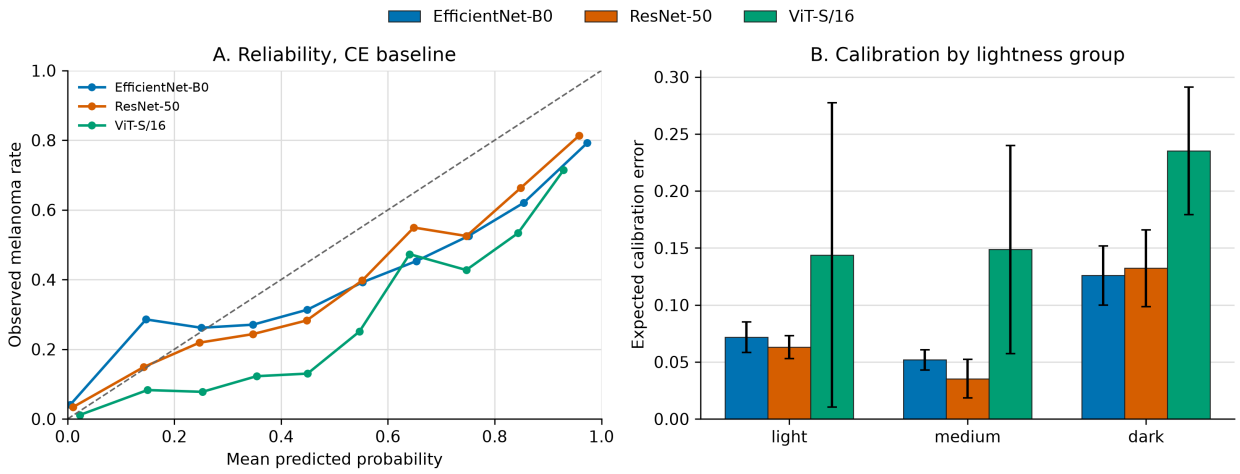


Figure 1: Calibration at the CE baseline. (A) Reliability curves for the three architectures. (B) Expected calibration error by lightness group, mean and seed standard deviation across the CE-baseline runs.

Table 6: Lightness-stratified test sensitivity at 95% specificity. Values are mean and standard deviation across three seeds. The dark stratum contains 43 test images and six melanoma cases. These are image-based lightness groups, not clinical Fitzpatrick labels.

Configuration	Light sens	Medium sens	Dark sens	Dark AUC	Dark FPR	Dark n / pos.
EfficientNet-B0 / CE	0.477 ± 0.031	0.609 ± 0.050	0.722 ± 0.096	0.871	0.117	43 / 6
EfficientNet-B0 / CE+strong	0.519 ± 0.054	0.599 ± 0.086	0.556 ± 0.096	0.833	0.090	43 / 6
EfficientNet-B0 / Focal+CB	0.420 ± 0.064	0.522 ± 0.076	0.500 ± 0.167	0.806	0.081	43 / 6
ResNet-50 / CE	0.502 ± 0.036	0.633 ± 0.050	0.444 ± 0.096	0.803	0.117	43 / 6
ResNet-50 / CE+strong	0.527 ± 0.026	0.603 ± 0.038	0.556 ± 0.096	0.854	0.135	43 / 6
ResNet-50 / Focal+CB	0.399 ± 0.093	0.481 ± 0.066	0.722 ± 0.192	0.848	0.162	43 / 6
ViT-S/16 / CE	0.342 ± 0.091	0.451 ± 0.065	0.333 ± 0.167	0.740	0.144	43 / 6
ViT-S/16 / CE+strong	0.407 ± 0.054	0.471 ± 0.132	0.222 ± 0.096	0.763	0.126	43 / 6
ViT-S/16 / Focal+CB	0.362 ± 0.062	0.586 ± 0.010	0.389 ± 0.096	0.712	0.189	43 / 6

at $L^* \leq 58$, and 0.073 at $L^* \leq 60$, against 0.047 to 0.050 for light and medium at every cutoff. The intervals around these group rates remain wide.

Dark sensitivity is 0.500 with 6 melanomas, 0.648 with 12, and 0.580 with 23. It is not consistently lower than light sensitivity, and the intervals are wide.

The adjusted models in Table 8 separate false positives from false negatives. The most robust fairness result is the false-negative association: the L^* odds ratio is 1.74 for ResNet-50, 1.66 for EfficientNet-B0, and 1.75 for ViT-S/16, and all three intervals exclude 1. Lighter skin is associated with more missed melanomas. The false-positive association points in the opposite direction, with L^* odds ratios of 0.78, 0.88, and 0.77, but the interval excludes 1 only for ResNet-50. Lesion contrast is associated with more false positives for all three architectures (OR 2.05, 1.69, and 2.42) and with fewer false negatives for ViT-S/16 (OR 0.48; the other two intervals include 1).

The error-type pattern is therefore suggestive but not symmetric. The supported finding is the false-negative lightness association; the false-positive association and the contrast effects are directionally consistent but imprecise. A single sensitivity or AUC summary hides this distinction.

Table 7: Dark-skin threshold sensitivity at 95% specificity. Values are means across the nine CE-baseline runs. The dark AUC gap is present at $L^* \leq 55$ but not at $L^* \leq 58$ or 60, while the dark false positive rate remains elevated at every cutoff. These are image-based lightness groups, not clinical Fitzpatrick labels.

L^* cutoff	Group	n	Melanomas	Sensitivity	AUC	FPR
55	light	644	81	0.440	0.868	0.050
	medium	840	99	0.565	0.911	0.049
	dark	43	6	0.500	0.805	0.126
58	light	644	81	0.440	0.868	0.050
	medium	777	93	0.550	0.908	0.048
	dark	106	12	0.648	0.891	0.086
60	light	644	81	0.440	0.868	0.050
	medium	704	82	0.556	0.912	0.047
	dark	179	23	0.580	0.881	0.073

Table 8: Adjusted odds ratios for false positives and false negatives. Each model includes perilesional L^*, lesion contrast, log lesion area, age, sex, and anatomical site. Odds ratios are per one standard deviation; intervals are lesion-cluster bootstrap 95% CIs. The pooled error model is not shown because the L^* effects on false positives and false negatives have opposite signs.

Error type	Architecture	L^* OR (95% CI)	Contrast OR (95% CI)
False positive	ResNet-50	0.78 (0.61 to 0.94)	2.05 (1.66 to 2.68)
	EfficientNet-B0	0.88 (0.67 to 1.11)	1.69 (1.39 to 2.14)
	ViT-S/16	0.77 (0.57 to 1.03)	2.42 (1.94 to 3.14)
False negative	ResNet-50	1.74 (1.11 to 2.58)	0.58 (0.35 to 1.02)
	EfficientNet-B0	1.66 (1.16 to 2.34)	0.74 (0.51 to 1.13)
	ViT-S/16	1.75 (1.14 to 2.60)	0.48 (0.29 to 0.75)

4.6 Quantitative XAI

Table 9 reports attribution mass inside the lesion, pointing-game accuracy, attribution IoU, and error rate for the CE-baseline models. Grad-CAM targets the true class, so a false-negative heatmap describes the evidence for melanoma rather than the benign prediction. Table 10 reports the Spearman correlations. The sample contains 129 dark images and 18 melanomas across the three seeds, and 450 light and 450 medium images.

Attribution mass inside the lesion is highest in the light stratum for all three architectures. EfficientNet-B0 falls from 0.450 to 0.368 to 0.331 across light, medium, and dark. ViT-S/16 falls from 0.341 to 0.260 and then stays near 0.261. ResNet-50 falls from 0.571 to 0.480 and then to 0.465. The mass-tone correlation is negative and significant for every architecture: -0.176 for EfficientNet-B0, -0.174 for ResNet-50, and -0.181 for ViT-S/16, all with $p < 10^{-7}$.

Pointing-game accuracy is lowest in the dark stratum in every case: 0.448 for EfficientNet-B0, 0.636 for ResNet-50, and 0.248 for ViT-S/16. Attribution IoU follows the same direction for EfficientNet-B0 and ViT-S/16, and is lower in the dark stratum than the medium stratum for ResNet-50 (0.301 vs 0.346).

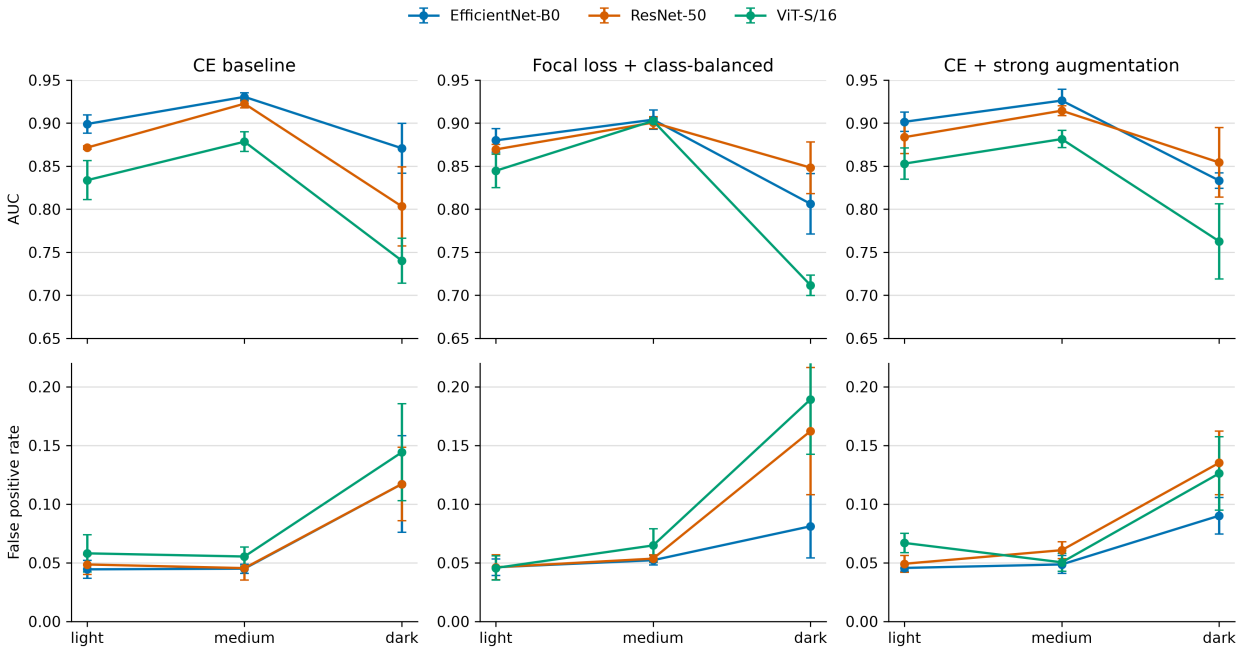


Figure 2: AUC and false positive rate by lightness group for the nine Stage 2 configurations at the pre-declared cutoff $L^* \leq 55$. Columns are training conditions and lines are architectures. Points are means across three seeds; error bars are seed standard deviations. The dark stratum contains 43 test images and six melanoma cases.

The mass-error correlation is architecture-dependent. It is negative for ResNet-50 (-0.230, $p < 10^{-13}$) and EfficientNet-B0 (-0.129, $p < 10^{-4}$), and positive for ViT-S/16 ($+0.136$, $p < 10^{-4}$). Higher attribution mass inside the lesion is associated with fewer errors for the CNNs and with more errors for the ViT. The false-negative and false-positive mass values do not separate consistently: false negatives vs false positives are 0.374 vs 0.260 for EfficientNet-B0, 0.346 vs 0.349 for ResNet-50, and 0.420 vs 0.344 for ViT-S/16. The XAI signal is a group-level tone gradient, not a consistent per-image error marker.

Figure 4 shows the group pattern. ResNet-50 localizes best in absolute terms and has the highest dark-stratum attribution mass, but its dark false positive rate is not the lowest. Localization quality and error rate move together across tone groups, but they do not rank architectures the same way.

4.7 External tone-labeled validation

Table 11 reports the HIBA dermoscopic subset for the CE-baseline models: 1,280 images and 196 melanomas. We applied the HAM10000 validation threshold at 95% specificity without refitting. HIBA contains Fitzpatrick I-IV only, and 108 images have no Fitzpatrick label.

Sensitivity drops on the external set. ResNet-50 reaches 0.391 ± 0.056 at 0.944 specificity, EfficientNet-B0 reaches 0.396 ± 0.036 at 0.922, and ViT-S/16 reaches 0.221 ± 0.130 at 0.967. The AUCs are 0.807 ± 0.018, 0.782 ± 0.024, and 0.791 ± 0.027. On the HAM10000 test set the same models reached 0.57, 0.56, and 0.40 sensitivity at 95% specificity. Both CNNs outperform the transformer on external sensitivity, but the two CNNs are within half a point of each other. The absolute drop is consistent with domain shift between the Argentine HIBA collection and HAM10000.

The per-type estimates do not show a monotonic skin-tone gradient. For Fitzpatrick I, II, and

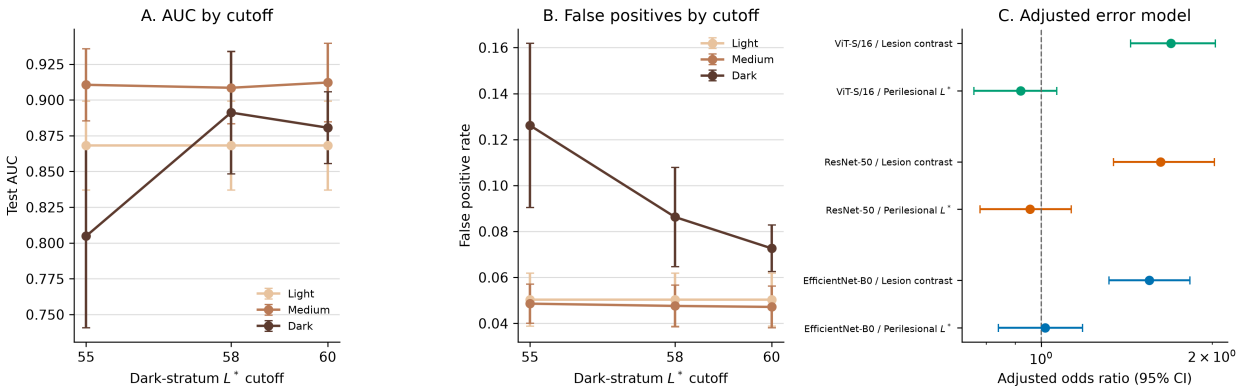


Figure 3: Dark-skin robustness. (A) AUC by L^* cutoff. (B) False positive rate by cutoff. (C) Adjusted odds ratios for the pooled error model, per standard deviation, with lesion-cluster bootstrap 95% CIs. The false-positive and false-negative models are reported separately in Table 8.

Table 9: Quantitative XAI at the CE baseline on the HAM10000 test split. Grad-CAM targets the true class. Each architecture uses three seeds and a tone-stratified sample of 343 test images per seed. Attribution mass is the fraction of the heatmap inside the lesion mask. Values are means across seeds.

Architecture	Tone group	n	n positive	Attribution mass	Pointing game	Attribution IoU	Error rate
ResNet-50	light	450	36	0.571	0.847	0.387	0.076
	medium	450	54	0.480	0.777	0.346	0.078
	dark	129	18	0.465	0.636	0.301	0.178
EfficientNet-B0	light	450	36	0.450	0.702	0.324	0.076
	medium	450	54	0.368	0.598	0.280	0.091
	dark	129	18	0.331	0.448	0.195	0.140
ViT-S/16	light	450	36	0.341	0.607	0.273	0.109
	medium	450	54	0.260	0.391	0.185	0.116
	dark	129	18	0.261	0.248	0.144	0.217

III, CNN sensitivity ranges from 0.35 to 0.43, and the ViT is lower at 0.20 to 0.33. Fitzpatrick IV contains 25 images and no melanomas, so sensitivity is undefined and only specificity can be estimated. The group sizes are unbalanced: 804 of the 1,280 images are type II. HIBA cannot test the dark-skin claim, because it has almost no type IV melanoma cases and no type V-VI images at all.

Table 12 compares all three conditions on HIBA. The focal condition raises ViT-S/16 sensitivity from 0.221 to 0.315, lowers ResNet-50 from 0.391 to 0.315, and lowers EfficientNet-B0 from 0.396 to 0.253. The ViT change is +0.219, +0.127, and −0.067 across seeds, with a mean of +0.094; the CNN changes are negative in every seed for EfficientNet-B0 and in two of three seeds for ResNet-50. The focal evidence is therefore directionally consistent for the CNNs and suggestive but not confirmatory for the ViT.

Strong augmentation is the best external condition for all three architectures. It raises sensitivity from 0.391 to 0.480 for ResNet-50, from 0.396 to 0.406 for EfficientNet-B0, and from 0.221 to 0.391 for ViT-S/16. The corresponding AUCs are 0.835, 0.847, and 0.824, and the oracle 95% specificity sensitivities are 0.408, 0.427, and 0.389. The near-neutral in-domain effect of strong augmentation therefore does not transfer: the condition designed to simulate domain shift improves external generalization. This is a more favorable external result for the Stage 1 winner than the in-domain

Table 10: Spearman correlations between attribution mass inside the lesion, lightness group, and error status at the CE baseline. Tone is coded light = 0, medium = 1, dark = 2. A negative mass-tone correlation means less attribution inside the lesion on darker images.

Architecture	Images	ρ (mass, tone)	p	ρ (mass, error)	p
EfficientNet-B0	1011	-0.176	1.9×10^{-8}	-0.129	4.2×10^{-5}
ResNet-50	1028	-0.174	2.0×10^{-8}	-0.230	9.1×10^{-14}
ViT-S/16	1029	-0.181	5.1×10^{-9}	0.136	1.2×10^{-5}

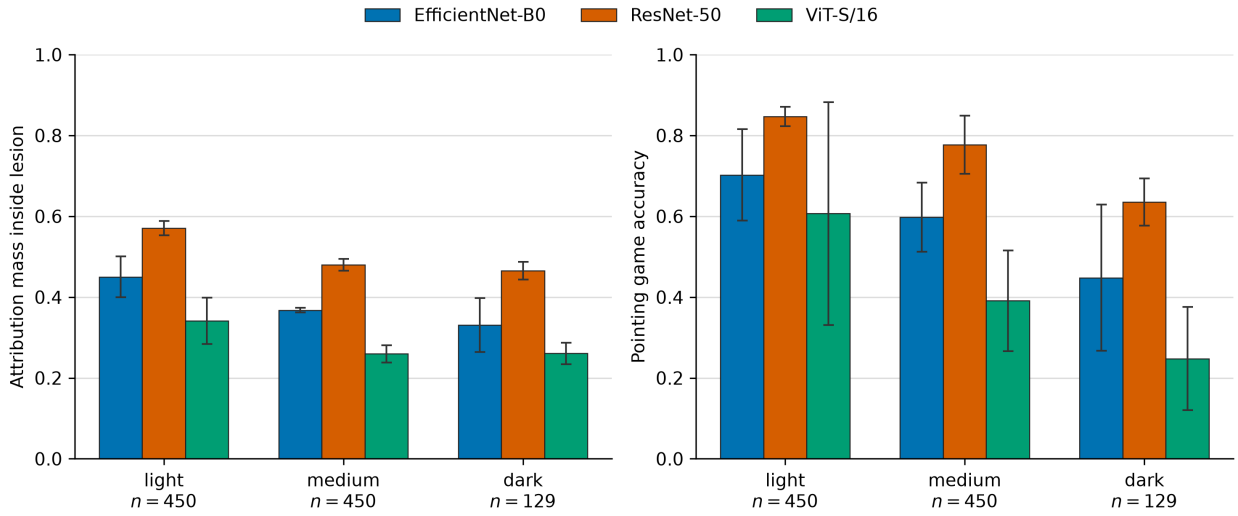


Figure 4: Attribution mass inside the lesion and pointing-game accuracy by lightness group for the three CE-baseline architectures. Grad-CAM targets the true class. Bars are means across three seeds; error bars are seed standard deviations. The sample contains all 43 dark test images per seed and 150 randomly drawn light and medium images per seed.

benchmark suggested.

At the oracle 95% specificity threshold, focal loss leaves the CNN sensitivity below the CE baseline and raises the ViT, while strong augmentation raises all three above the CE baseline. The external AUC ordering also differs from the in-domain ordering. Under focal loss the external AUC ranking is ViT-S/16 0.794, ResNet-50 0.774, and EfficientNet-B0 0.722, while the in-domain focal ranking is EfficientNet-B0 0.891, ResNet-50 0.884, and ViT-S/16 0.872. HIBA is a different imaging regime, not a clean replication.

The external validation supports a limited conclusion. Strong augmentation generalizes best, the CNN sensitivity loss under focal loss is consistent in direction, and the ViT focal gain is possible but seed-dependent. The absolute performance of the CE baseline does not transfer, so the operating point needs recalibration for a new source. The tone-specific claim remains limited to the image-based lightness strata in HAM10000 and to the error-type associations reported above.

Table 11: External validation on the HIBA dermoscopic subset for the CE-baseline models (1,280 images, 196 melanomas). The threshold is the HAM10000 validation threshold at 95% specificity, frozen before the external evaluation. Values are mean and standard deviation across three seeds. Fitzpatrick IV contains 25 images and no melanomas, so sensitivity is undefined. The 108 images without a Fitzpatrick label are included in the All row only.

Architecture	Fitzpatrick	n	Melanomas	Sensitivity	Specificity	AUC
ResNet-50	I	97	25	0.360 ± 0.069	0.954 ± 0.021	0.679 ± 0.031
	II	804	144	0.403 ± 0.066	0.922 ± 0.011	0.780 ± 0.020
	III	246	17	0.412 ± 0.000	0.978 ± 0.004	0.946 ± 0.011
	IV	25	0	–	1.000 ± 0.000	–
	All	1280	196	0.391 ± 0.056	0.944 ± 0.006	0.807 ± 0.018
EfficientNet-B0	I	97	25	0.427 ± 0.061	0.940 ± 0.021	0.741 ± 0.068
	II	804	144	0.389 ± 0.039	0.890 ± 0.010	0.745 ± 0.028
	III	246	17	0.353 ± 0.059	0.974 ± 0.004	0.860 ± 0.044
	IV	25	0	–	0.987 ± 0.023	–
	All	1280	196	0.396 ± 0.036	0.922 ± 0.008	0.782 ± 0.024
ViT-S/16	I	97	25	0.200 ± 0.144	0.995 ± 0.008	0.773 ± 0.061
	II	804	144	0.227 ± 0.131	0.950 ± 0.033	0.777 ± 0.018
	III	246	17	0.333 ± 0.189	0.994 ± 0.007	0.849 ± 0.053
	IV	25	0	–	1.000 ± 0.000	–
	All	1280	196	0.221 ± 0.129	0.967 ± 0.022	0.791 ± 0.027

5 Discussion

5.1 Imbalance correction is architecture-dependent

Focal loss with class-balanced sampling is not a neutral intervention. It raises the transformer and lowers both CNNs in the point estimates. A benchmark that applied the correction to one architecture and generalized the result would have reached the wrong conclusion about the others.

The seed-level tests sharpen this claim and limit it. Every ResNet-50 and EfficientNet-B0 seed falls under focal loss, and every ViT-S/16 seed rises, but the within-architecture 95% t intervals all include zero. The difference-in-differences is significant for EfficientNet-B0 against ViT-S/16, but not for ResNet-50 against ViT-S/16. The supported finding is therefore an architecture-dependent interaction with limited precision, driven by the EfficientNet-B0 versus ViT contrast. The point estimates reverse, but three seeds do not support a narrow effect size or a definitive ordering.

The effect has a plausible mechanism. Focal loss changes the gradient distribution, and class-balanced sampling changes the effective class prior. A transformer with weaker inductive bias may benefit from the harder examples and the reweighted prior, while a CNN with strong local features may overfit the minority samples or lose calibration. The experiment cannot separate those mechanisms. We report the effect as an empirical interaction, not a causal explanation.

5.2 Calibration and AUC both move

The sensitivity ranking reverses, but the AUC ordering does not. EfficientNet-B0 remains first and ViT-S/16 remains last in both conditions. The gaps, however, close: EfficientNet-B0 exceeds ViT-S/16 by 0.060 at the CE baseline and by only 0.018 under focal loss, and the focal AUC differences are not significant.

Table 12: External validation on HIBA for all three training conditions. Thresholds are the frozen HAM10000 validation thresholds. Sensitivity at 95% specificity uses an oracle threshold fitted on HIBA and is an upper bound, not a deployable result. Values are mean and standard deviation across three seeds.

Architecture	Condition	Sensitivity	Specificity	AUC	Sens. at 95% spec.
ResNet-50	CE	0.391 ± 0.056	0.944 ± 0.006	0.807 ± 0.018	0.366 ± 0.036
ResNet-50	CE+strong	0.480 ± 0.032	0.935 ± 0.011	0.835 ± 0.010	0.408 ± 0.044
ResNet-50	Focal+CB	0.315 ± 0.070	0.925 ± 0.005	0.774 ± 0.017	0.245 ± 0.057
EfficientNet-B0	CE	0.396 ± 0.036	0.922 ± 0.008	0.782 ± 0.024	0.311 ± 0.035
EfficientNet-B0	CE+strong	0.406 ± 0.069	0.955 ± 0.014	0.847 ± 0.003	0.427 ± 0.024
EfficientNet-B0	Focal+CB	0.253 ± 0.071	0.945 ± 0.018	0.722 ± 0.045	0.240 ± 0.040
ViT-S/16	CE	0.221 ± 0.129	0.967 ± 0.022	0.791 ± 0.027	0.310 ± 0.084
ViT-S/16	CE+strong	0.391 ± 0.108	0.940 ± 0.023	0.824 ± 0.018	0.389 ± 0.023
ViT-S/16	Focal+CB	0.315 ± 0.052	0.954 ± 0.011	0.794 ± 0.011	0.335 ± 0.016

The calibration results explain part of the pattern. Focal loss raises expected calibration error by 4.4x for ResNet-50 and 3.0x for EfficientNet-B0, but only 1.36x for ViT-S/16. Calibration degradation therefore tracks the CNN loss. It does not explain the ViT gain, which comes with a small AUC improvement. The ViT gain is a ranking improvement with a modest calibration change, while the CNN loss is a mixture of ranking loss and calibration degradation.

This matters for how model comparisons are reported. A single threshold at 0.5, or a single sensitivity number, can make an architecture look better or worse depending on the correction and the calibration. Reporting sensitivity at a fixed specificity, AUC, calibration, and seed-level intervals together gives a more complete picture. The calibration analysis should be part of the model selection record, not an afterthought.

5.3 The fairness signal is error-type-specific and imprecise

At the pre-declared cutoff $L^* \leq 55$, the dark stratum has the lowest AUC and the highest false positive rate in all nine configurations. The AUC result is not statistically supported: none of the 54 dark-versus-light and dark-versus-medium AUC comparisons across the three cutoffs has an interval that excludes zero. The false positive elevation is more consistent. The dark false positive rate is 0.126 at $L^* \leq 55$, 0.086 at $L^* \leq 58$, and 0.073 at $L^* \leq 60$, against 0.047 to 0.050 for light and medium at every cutoff. The dark sensitivity estimate is not consistently lower than light sensitivity, and it is too uncertain to interpret.

The adjusted models give the clearest result. Lighter skin is associated with more false negatives: the L^* odds ratio is 1.74, 1.66, and 1.75 for the three architectures, and all intervals exclude 1. Darker skin is associated with more false positives in the point estimates, but the interval excludes 1 only for ResNet-50. Lesion contrast is associated with more false positives for all three architectures and with fewer false negatives for ViT-S/16.

This is a more precise fairness claim than a global sensitivity gap, but it is also a weaker one than a confirmed dark-skin AUC deficit. The model does not simply perform worse on darker skin. The error profile shifts: lighter images produce more missed melanomas, and darker images produce more false alarms. The false-negative association is the robust part. The false-positive association and the contrast effects are directionally consistent but imprecise.

The lesion-contrast result also changes the interpretation of the light stratum. Pale lesions on pale skin have low contrast, and the model is more likely to miss them. Dark lesions on lighter skin

have high contrast, and the model is more likely to call them malignant even when they are benign. Skin tone and lesion contrast are correlated, but the adjusted models show that the error-type shift is better described by the lesion-skin contrast than by absolute lightness alone.

5.4 The data bottleneck

The external dermoscopic datasets with Fitzpatrick labels, HIBA and PROVe-AI, contain Fitzpatrick I-IV only. The HIBA evaluation makes the constraint concrete. In 1,280 dermoscopic images, Fitzpatrick IV accounts for 25 images and not a single melanoma. Sensitivity is estimable only for types I-III, and those estimates show no monotonic skin-tone gradient.

The external condition comparison is directionally consistent for the CNNs and suggestive for the ViT. Focal loss lowers EfficientNet-B0 sensitivity in every seed and ResNet-50 in two of three seeds. The ViT changes are +0.219, +0.127, and -0.067, so the mean gain hides a negative seed. The external evidence is therefore not a clean replication of the in-domain interaction. Strong augmentation is the best external condition for all three architectures, which supports its use as a domain-shift simulation even though it was neutral in-domain.

The clinical datasets with Fitzpatrick V-VI, such as DDI, SCIN, and PASSION, differ from HAM10000 in imaging modality. This is the main obstacle to a strong dark-skin claim. This study can quantify the representation gap, report the persistent false-positive elevation and the false-negative lightness association, and show the limitation of the available dermoscopic data. A precise dark-skin sensitivity estimate requires new dermoscopic data collection or a carefully controlled cross-modality study.

5.5 What XAI adds

The XAI results show a group-level pattern that the aggregate metrics do not capture. All three architectures place less attribution inside the lesion on darker images, with mass-tone correlations of -0.17 to -0.18. The direction is consistent across a standard CNN, an efficient CNN, and a transformer, so it is not a quirk of one inductive bias.

The mass-error relationship is architecture-dependent. Higher attribution mass inside the lesion is associated with fewer errors for ResNet-50 and EfficientNet-B0, and with more errors for ViT-S/16. The false-negative and false-positive mass values do not separate consistently. With the true class as the Grad-CAM target, the earlier apparent separation disappears. Attribution mass is a descriptive group-level signal, not a consistent per-image error marker.

This sets a limit on what attribution can claim. A deployment system should not use this heatmap score as a per-image triage signal without further work. It can use the group-level pattern as a monitoring signal: when a cohort shows lower attribution mass inside the lesion, the model may be attending less to the lesion in that cohort. The direction of the change matters more than the absolute value.

ResNet-50 is the counterexample to a simple assumption that more attribution inside the lesion means fewer errors. It has the highest attribution mass and pointing-game accuracy in the dark stratum, but its dark false positive rate is not the lowest. Heatmaps are a partial view of model behaviour, not a substitute for group-level performance measurement.

5.6 Implications for model selection

The choice of architecture depends on the correction, but the seed-level intervals are wide. At the CE baseline, ResNet-50 has the highest point-estimate sensitivity at 95% specificity. Under focal loss with class-balanced sampling, ViT-S/16 has the highest point estimate. A deployment decision

that ignores the training condition can pick a different model, but the current evidence supports an architecture-dependent effect rather than a precise ranking.

For the fairness analysis, the evaluation should include the architecture-condition combinations, the calibration, and the two error types. The best model on aggregate sensitivity may not have the lowest false positive rate in the dark stratum, and a model with a high AUC may still shift its errors in a way that matters clinically. The external validation adds a second source, but the available Fitzpatrick labels do not reach the darkest groups.

6 Limitations

- The training data are HAM10000 only. HIBA adds one external dermoscopic source, but cross-dataset generalization to PH2, BCN20000 [10], or ISIC 2020 is not measured.
- The endpoint is binary melanoma versus all other diagnoses. Seven-class classification was not evaluated.
- Each condition uses three seeds. Every within-architecture 95% t interval includes zero, and the seed-level intervals are wide. The paper supports an architecture-dependent interaction with limited precision, not a precise ranking.
- The AUC ordering is preserved in the point estimates, but the focal AUC differences are not significant and the ViT closes the gap. The sensitivity reversal is therefore not purely a threshold effect.
- The external focal validation shows a consistent CNN loss and a seed-dependent ViT gain. The ViT external change is +0.219, +0.127, and -0.067 across seeds.
- Stage 1 uses 160-pixel images and Stage 2 uses 224-pixel images. The two stages are not directly comparable.
- The fairness analysis uses an image-based lightness proxy, not clinical Fitzpatrick labels. The proxy is sensitive to illumination, color calibration, and lesion contrast.
- The pre-declared dark stratum contains 43 test images and six melanomas. The cutoff sensitivity analysis adds groups with 12 and 23 melanomas, but all dark-stratum estimates remain wide.
- The adjusted false-positive and false-negative models are observational. They identify associations, not causal effects, and the false-negative model has only 186 melanoma images across three seeds.
- HIBA and PROVe-AI contain Fitzpatrick I-IV only. HIBA has no Fitzpatrick IV melanomas, and the clinical datasets with Fitzpatrick V-VI differ in imaging modality, so the dark-skin comparison is confounded.
- Deletion and insertion curves saturated at a target probability near 1.0, so they could not separate heatmaps. Faithfulness is therefore not measured, and the XAI results cover localization only. Grad-CAM targets the true class, but the false-negative and false-positive mass values do not separate consistently. A second attribution method is not included.
- No clinician reader study was conducted, so the clinical relevance of the attribution maps is not validated with dermatologists.

- No deployed system, prospective validation, or regulatory analysis is included.

7 Ethics and reproducibility

All data are public, de-identified research datasets. HAM10000 and the ISIC archive are released for research use, and the external datasets are used under their respective licenses. The study involves no patient contact and no new data collection.

The system is a research prototype, not a diagnostic device. The results do not establish clinical utility.

The Fitzpatrick scale has known limitations: it is a coarse phototype scale, it does not capture the full range of skin tones, and it conflates pigmentation with UV response. We use it because the datasets and the fairness literature use it. The lightness strata are image-based measurements, not clinical labels.

The representation gap is a finding with an ethical consequence: the data needed to measure dark-skin performance are scarce. We report the small number of dark-skin dermoscopic images. Synthetic dark-skin images are not used to claim fairness improvements.

The code, configs, splits, per-sample predictions, and table generation scripts are released with the paper. The repository records the git commit, seed, device, and package versions for every run.

Author contributions

N.Z. designed the study, implemented the code, ran the experiments, and wrote the manuscript. S.B. supervised the project and revised the manuscript. Both authors have read and agreed to the published version of the manuscript.

Institutional review board statement

Not applicable. All data are public, de-identified research datasets, and the study involved no patient contact or new data collection.

Informed consent statement

Not applicable.

Data availability

HAM10000 is available from the ISIC archive. HIBA is available from the ISIC archive under a CC-BY license. The per-sample test predictions, the lesion-aware split file, the contrast features, fairness summaries, XAI metrics, and external-validation results are released with the code repository.

Code availability

The code is at `https://github.com/conorkz/melanoma-xai-bias`. The repository contains the training configs, split construction, evaluation scripts, table generators, and tests. The results in this paper were generated from the committed version of that repository. The submission version is tagged `v1.0-submission`.

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Conflicts of interest

The authors declare no conflicts of interest.

8 Conclusion

We benchmarked ResNet-50, EfficientNet-B0, and ViT-S/16 on HAM10000 under three pre-declared conditions. Focal loss with class-balanced sampling raised ViT-S/16 sensitivity by 8.2 points and lowered the two CNNs, which reversed the point-estimate ranking at 95% specificity. The direction was consistent across seeds, but every within-architecture 95% t interval included zero. The EfficientNet-B0 versus ViT interaction was significant at the seed level; the ResNet-50 versus ViT interaction was not. The AUC ordering was preserved, but the gaps closed under focal loss. Calibration error rose 4.4x for ResNet-50, 3.0x for EfficientNet-B0, and 1.36x for ViT-S/16, so calibration tracked the CNN loss rather than the ViT gain.

The fairness analysis became more specific once the cutoff and the error type were separated. The dark stratum had a higher false positive rate at L^* cutoffs of 55, 58, and 60, but none of the dark-versus-light or dark-versus-medium AUC differences was significant. The false-negative L^* association was significant for all three architectures; the false-positive association was significant only for ResNet-50. Fairness in this setting is an error-type shift with limited precision, not a confirmed dark-skin AUC deficit.

The XAI analysis showed that all three architectures place less attribution inside the lesion on darker images. The mass-error relationship differed by architecture, and the false-negative versus false-positive mass values did not separate consistently. HIBA validation showed a consistent CNN sensitivity loss under focal loss, a seed-dependent ViT gain, and the best external performance under strong augmentation. HIBA contains no Fitzpatrick IV melanomas and no Fitzpatrick V-VI images.

The next steps are a clinical dark-skin evaluation with DDI or PASSION, additional seeds for the six focal and CE cells, and a second attribution method. The clinical dataset is the main requirement for a precise Fitzpatrick V-VI claim. The additional seeds would tighten the interaction, and the second attribution method would test whether the tone gradient is specific to Grad-CAM.

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